Exp no:6 Determination and plotting of Capacitance of parallel plate capacitor in three different dielectric mediums with specific area, dielectric thickness and various dielectric constant.

Program:

exp333.sci (C:\Users\pavan\Documents\beee\exp...

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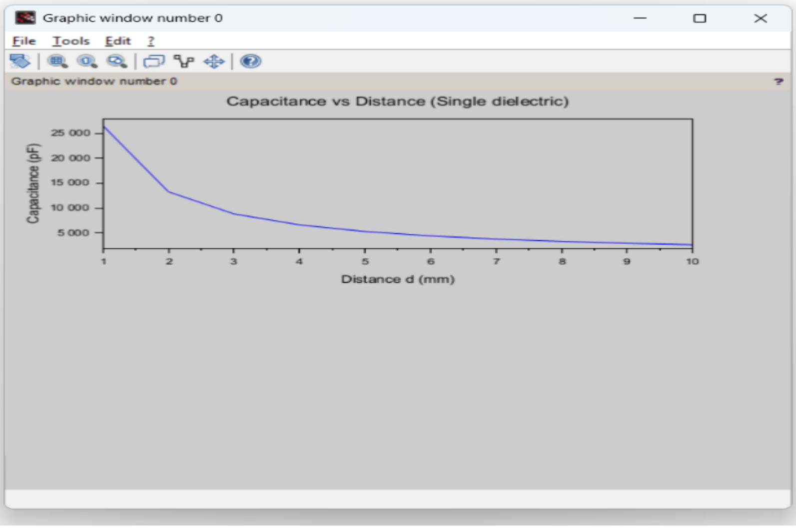
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes

*exp333.sci

```
clc;
clear;
e = 8.854e-12;
a1 = input("Enter the a1 value (m^2):");
d  = input("Enter the d value (m):");
c1 = e * a1 / d;
disp("Capacitance (single dielectric):");
disp(c1);
a2  = input("Enter the a2 value (m^2):");
d1  = input("Enter the d1 value (m):");
d2  = input("Enter the d2 value (m):");
d3  = input("Enter the d3 value (m):");
er1 = input("Enter the er1 value:");
er2 = input("Enter the er2 value:");
er3 = input("Enter the er3 value:");
c2 = (e * a2) / ( (d1/er1) + (d2/er2) + (d3/er3) );
disp("Capacitance (layered dielectric):");
disp(c2);
d_values = 0.001:0.001:0.01; // vary distance from 1 mm to 10 mm
C_values = (e * a1) ./ d_values;
er1_values = 1:1:10;
C_layered = (e * a2) ./ ( (d1 ./ er1_values) + (d2/er2) + (d3/er3) );
figure();
subplot(2,1,1);
plot(d_values*1000, C_values*1e12, 'b');
xlabel("Distance d (mm)");
ylabel("Capacitance (pF)");
title("Capacitance vs Distance (Single dielectric)");
grid();
subplot(2,1,2);
plot(er1_values, C_layered*1e12, 'r');
xlabel("Dielectric constant er1");
ylabel("Capacitance (pF)");
title("Capacitance vs er1 (Layered dielectric)");
grid();
```

Output:

```
Enter the a1 value (m^2):3

Enter the d value (m):1

  "Capacitance (single dielectric):"
   2.656D-11
Enter the a2 value (m^2):5

Enter the d1 value (m):2

Enter the d2 value (m):4

Enter the d3 value (m):6

Enter the er1 value:2

Enter the er2 value:3

Enter the er3 value:4

  "Capacitance (layered dielectric):"
   1.155D-11
Undefined variable: grid

-->
```



Calculation:

Exp6 :-

$a_1 = 3\ cm^2$

$a_2 = 5\ cm^2$

$d = 1\ mm$

$d_1 = 2\ mm$

$d_2 = 4\ mm$

$d_3 = 6\ mm$

$\varepsilon r_1 = 2$

$\varepsilon r_2 = 3$

$\varepsilon r_3 = 4$

A)(i) Single dielectric medium.

$$C = \varepsilon_0 \cdot \frac{a_1}{d} \qquad \varepsilon_0 = 8.854 \times 10^{-12}\ F/m.$$

$$C = \frac{8.854 \times 10^{-13} \times 3}{5} = 5.312 \times 10^{-12}\ F$$

(ii) three dielectric layers in series.

O/P

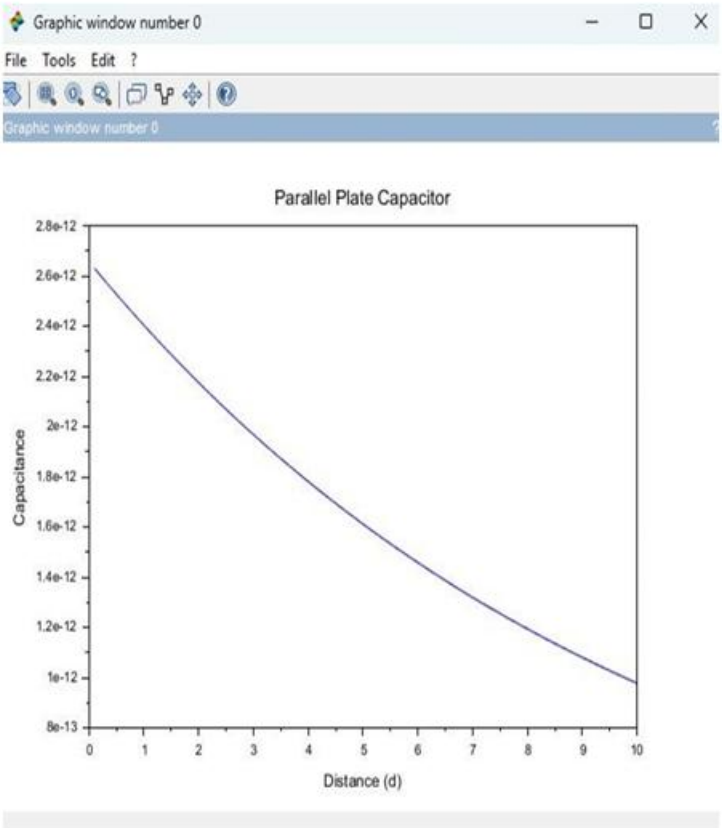
$$C = \frac{\varepsilon_0 \cdot a_2}{\frac{d_1}{\varepsilon_{r1}} + \frac{d_2}{\varepsilon_{r2}} + \frac{d_3}{\varepsilon_{r3}}}$$

$$C = \frac{8.854 \times 10^{-12} \times 1}{\frac{2}{2} + \frac{4}{3} + \frac{6}{4}} = \frac{8.854 \times 10^{-12}}{3.833} = 2.310 \times 10^{-12}\ F$$

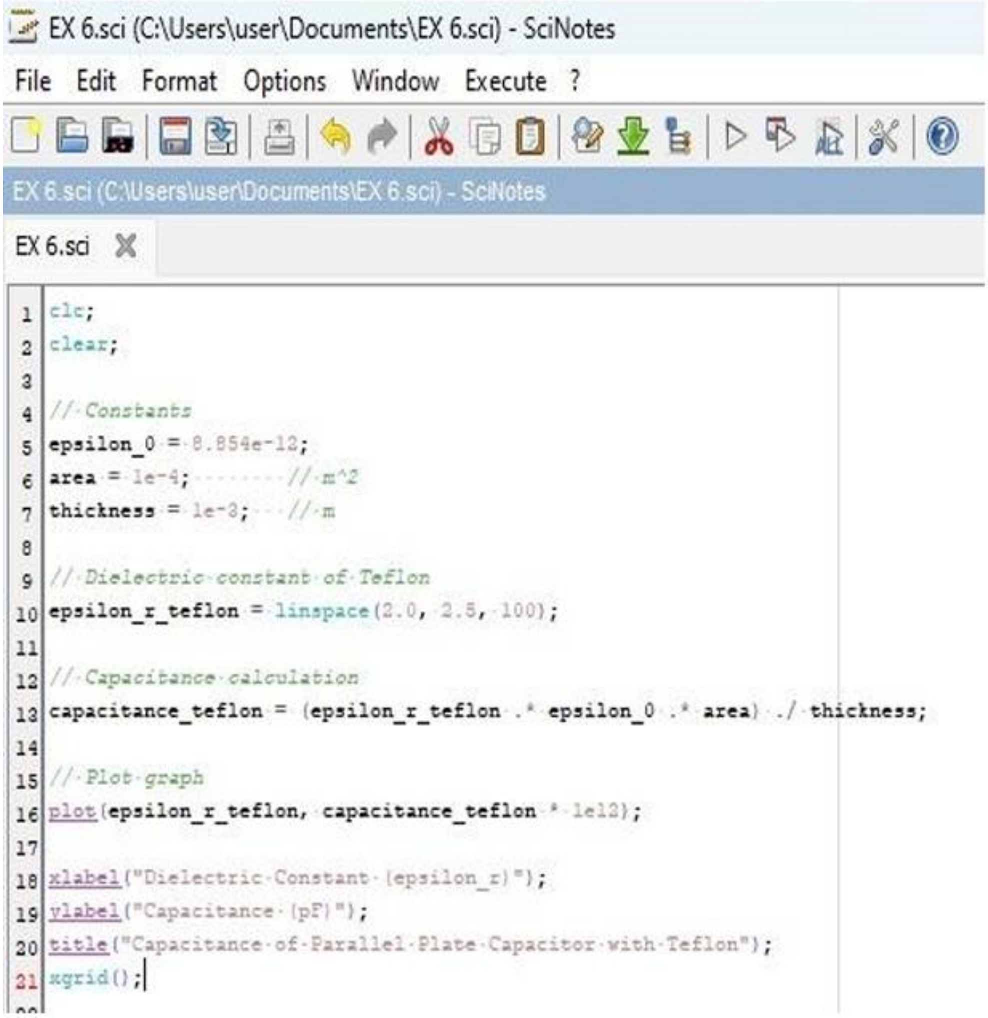
Case 1:

```
epsilon_0 = 8.854e-12;
area = 1e-4;
thickness = 1e-3;
epsilon_r_air = linspace(1.0, 1.5, 100);
capacitance_air = (epsilon_r_air .* epsilon_0 .* area) ./
thickness;
plot(epsilon_r_air, capacitance_air * 1e12, 'b');
xlabel("Dielectric Constant (\epsilon_r)");
ylabel("Capacitance (pF)");
title("Capacitance of Parallel Plate Capacitor with Air");

xtitle("Capacitance vs Dielectric Constant for Air");
grid();
```

Output:

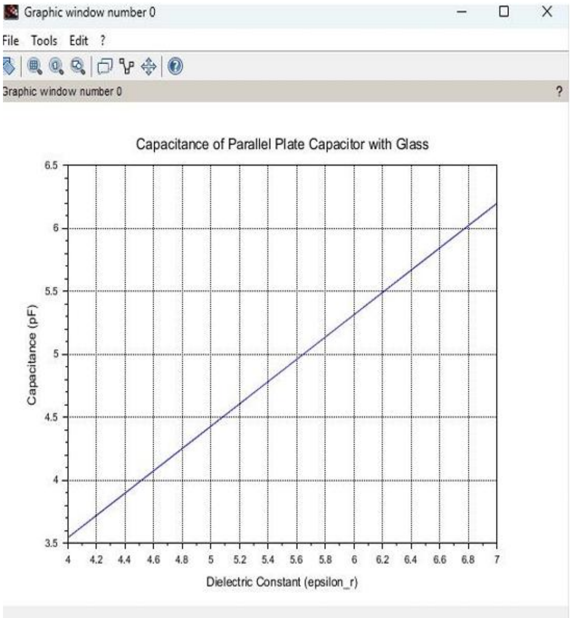


Case 2. Glass:




```
clc;
clear;

// Constants
epsilon_0 = 8.854e-12;
area = 1e-4;        // m^2
thickness = 1e-3;   // m

// Dielectric constant of Teflon
epsilon_r_teflon = linspace(2.0, 2.5, 100);

// Capacitance calculation
capacitance_teflon = (epsilon_r_teflon .* epsilon_0 .* area) ./ thickness;

// Plot graph
plot(epsilon_r_teflon, capacitance_teflon * 1e12);

xlabel("Dielectric Constant (epsilon_r)");
ylabel("Capacitance (pF)");
title("Capacitance of Parallel Plate Capacitor with Teflon");
xgrid();
```

Output:



CASE:3

EX 6.sci (C:\Users\user\Documents\EX 6.sci) - SciNotes

```
clc;
clear;

// Constants
epsilon_0 = 8.854e-12;
area = 1e-4;        // m^2
thickness = 1e-3;   // m

// Dielectric constant of glass
epsilon_r_glass = linspace(4.0, 7.0, 100);

// Capacitance calculation
capacitance_glass = (epsilon_r_glass .* epsilon_0 .* area) ./ thickness;

// Plot
plot(epsilon_r_glass, capacitance_glass * 1e12);

xlabel("Dielectric Constant (epsilon_r)");
ylabel("Capacitance (pF)");
title("Capacitance of Parallel Plate Capacitor with Glass");
xgrid();
```

Output: